Quantum Sensing hands-on session

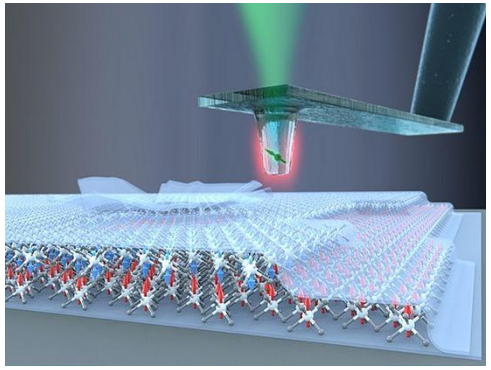
12th October 2026

August Gude

Quantum Control for Fundamental Physics

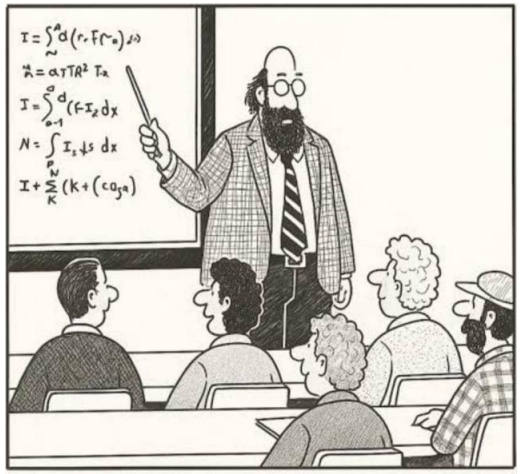
Quantum sensing

- Quantum systems are great at sensing environmental effects
  - NV center magnetometry
  - Atom interferometry



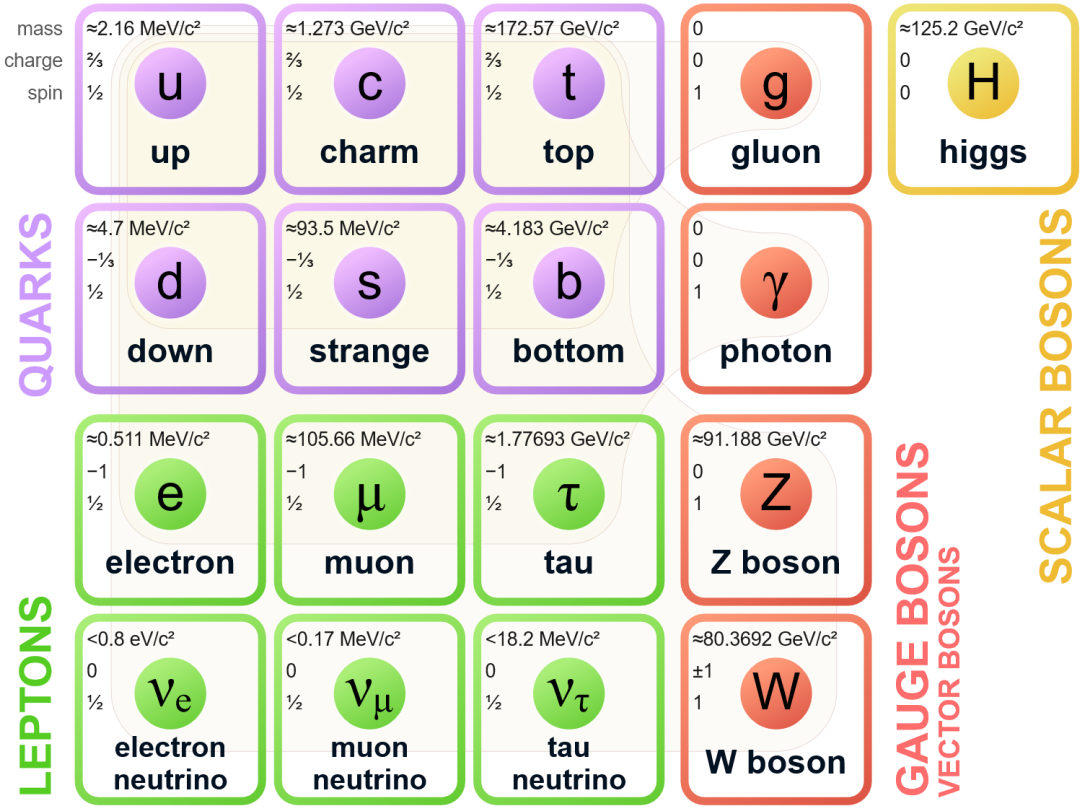
- But also very sensitive probes of fundamental physics
  - LIGO: Gravitational waves
  - Standard Model tests
  - Probe Beyond-Standard-Model physics

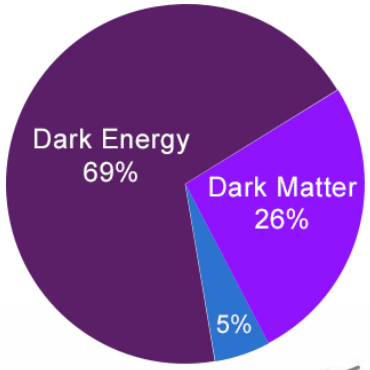




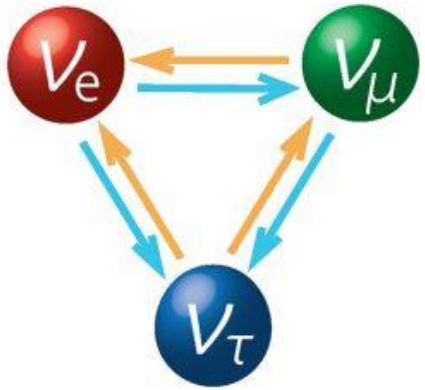
"Along with 'Antimatter,' and 'Dark Matter,' we've recently discovered the existence of 'Doesn't Matter,' which appears to have no effect on the universe whatsoever."

The Standard Model



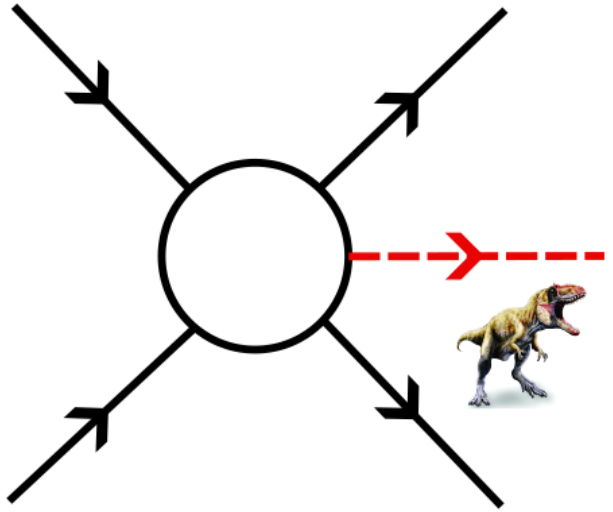






Beyond-Standard-Model physics searches

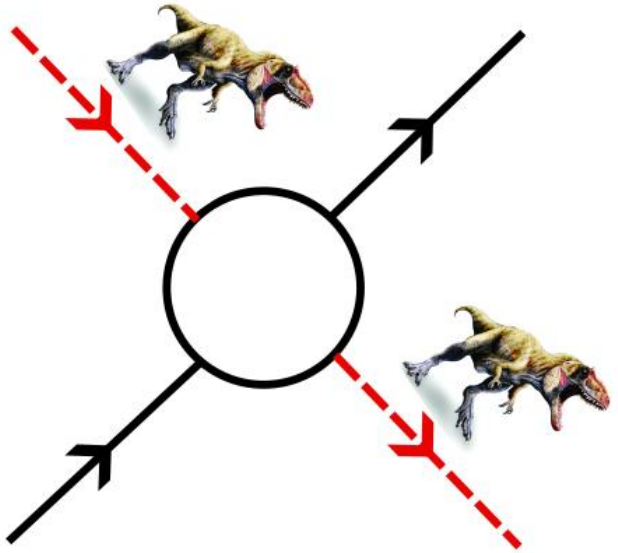
Energy frontier



Collider experiments:

- LHC
- SLAC
- …

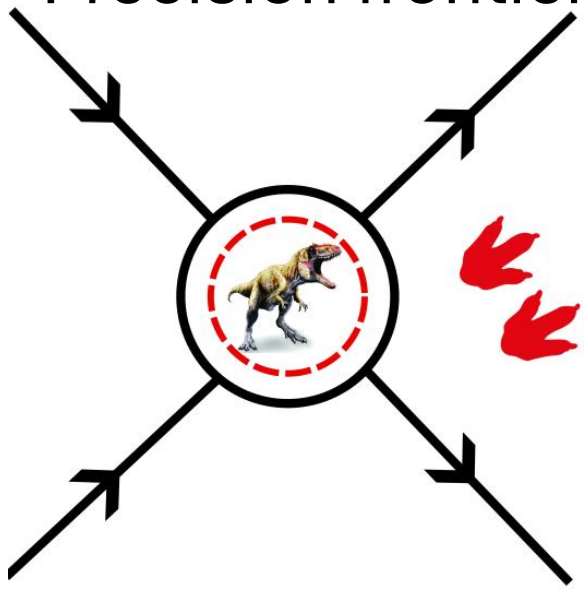
Cosmic frontier



Underground detectors:

- LUX-ZEPLIN
- SNOLAB
- …

Precision frontier

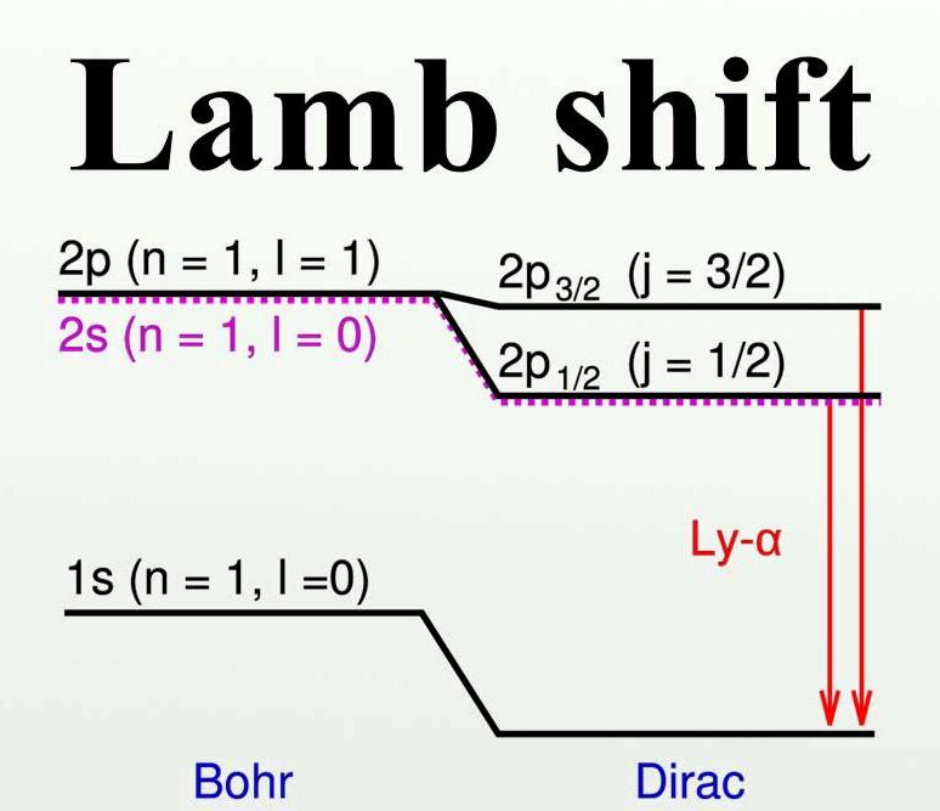


Tabletop AMO experiments:

- eEDM
- APV
- …

Precision frontier in atomic systems

1. Measure an atomic observable
2. Compare to the theoretical value
3. Look for deviations
4. **Improve the theory** and the experiments
5. Repeat with increased precision



Precision frontier in atomic systems

- Which atomic properties to measure?
  - "Never measure anything but frequency" – Arthur Schawlow

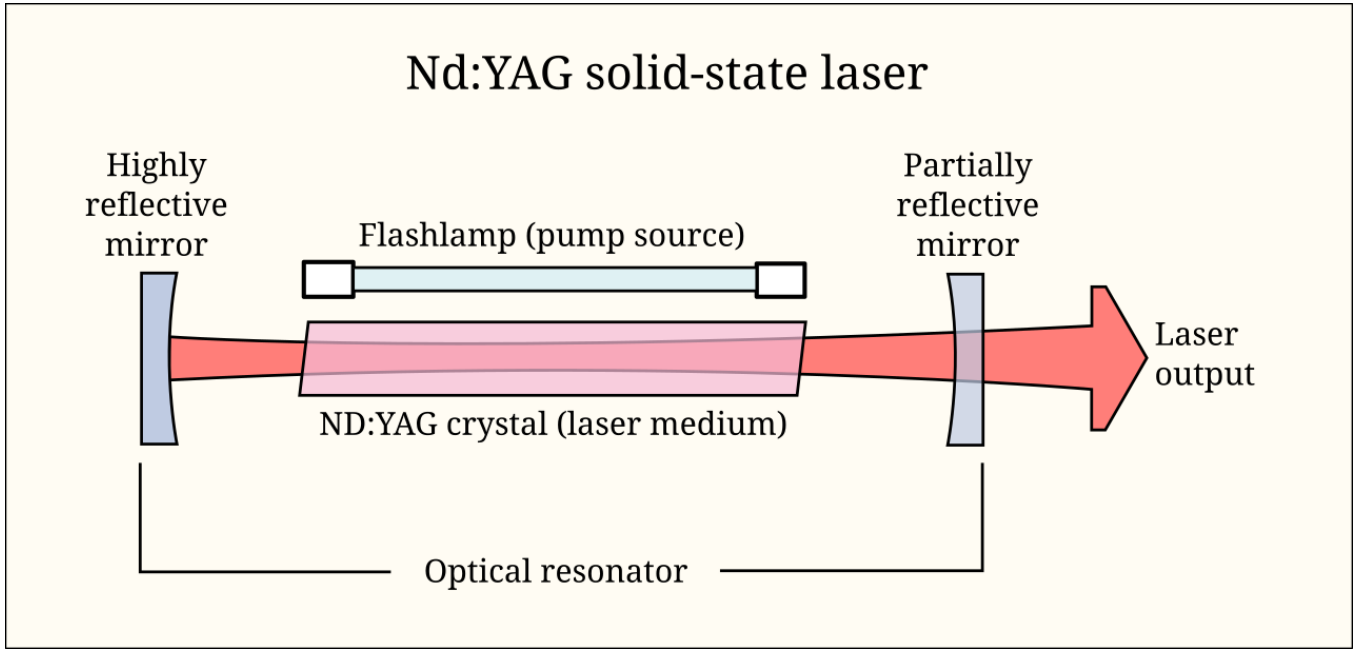
Example
  - Electron's magnetic moment (fractional uncertainty in the 14th digit)
  - Hydrogen 1S-2S transition frequency (fractional uncertainty in the 15th digit)
  - …

- Fundamental requirement: Stable frequency reference!
  - In the optical domain: Laser

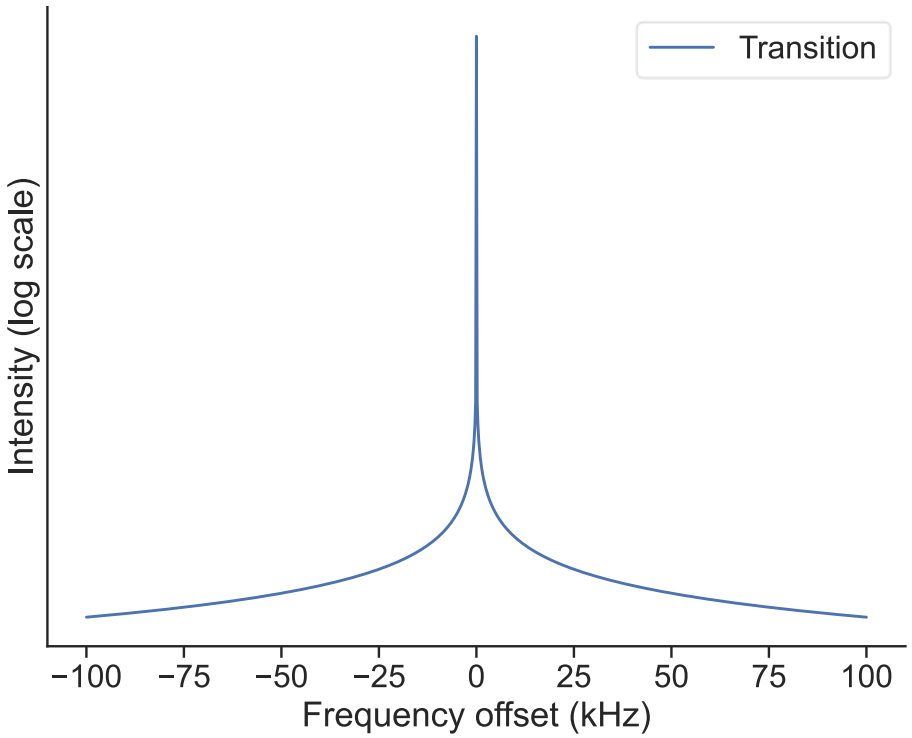
Laser

- Monochromatic source of light?

- Output frequency depends on
  - Gain material
  - Pump process of the gain material
  - Length of the cavity
  - …

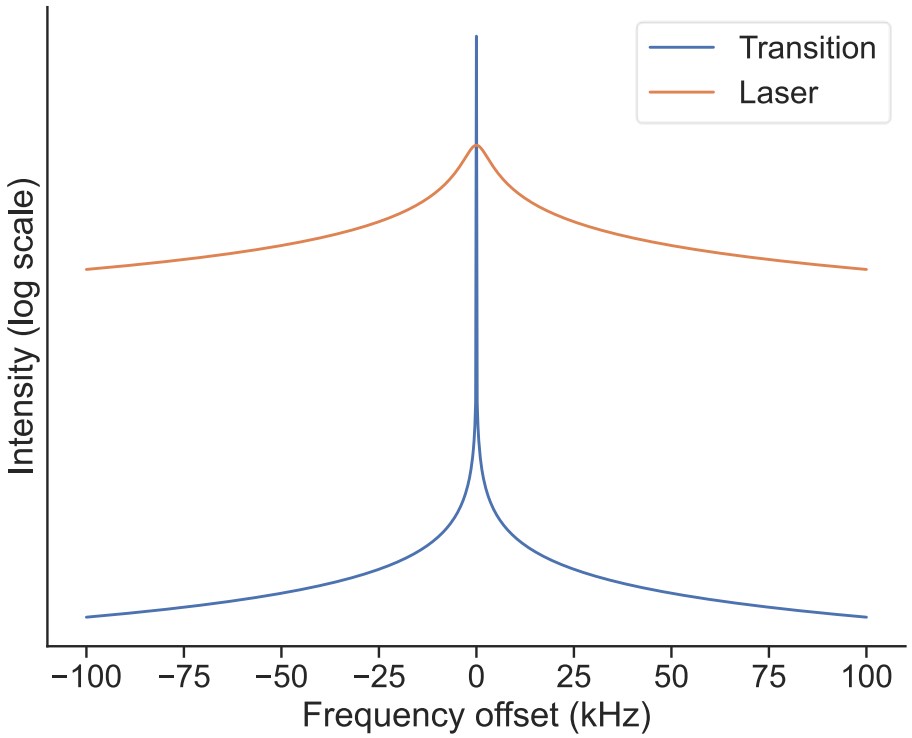


Laser frequency and linewidth



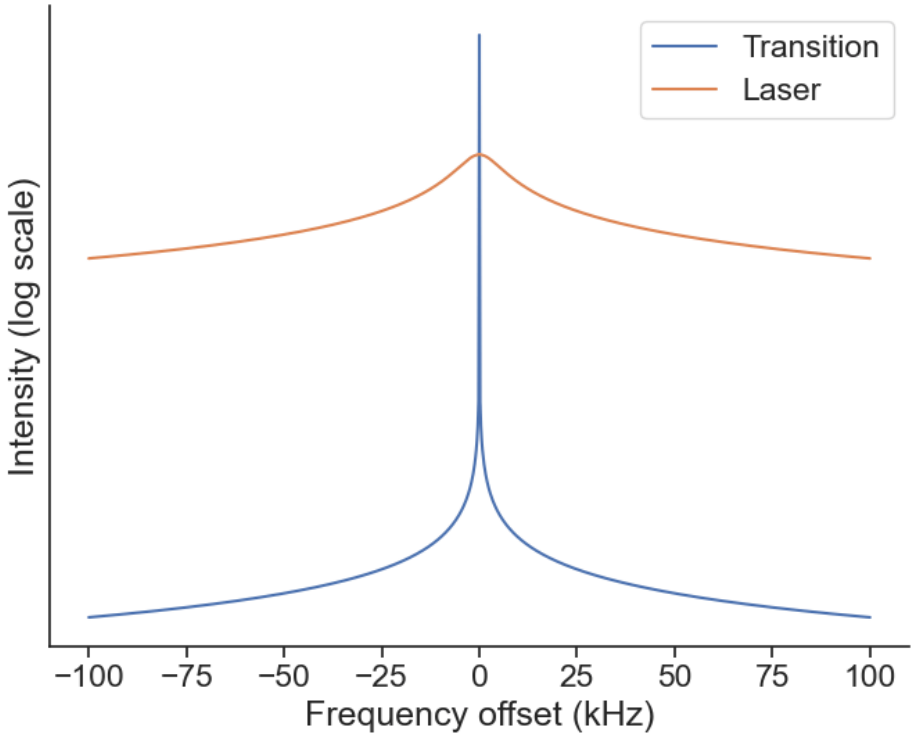
- Many atomic transitions are very narrow
- Need a narrow and stable laser to excite the transition

Laser frequency and linewidth



- Many atomic transitions are very narrow
- Need a narrow and stable laser to excite the transition

Laser frequency and linewidth



- Many atomic transitions are very narrow
- Need a narrow and stable laser to excite the transition

Not just for precision experiments!

- Narrow transitions are also used to define qubits in atoms and ions
  - High-fidelity operations require stable lasers!
- Laser stabilization is also required for auxiliary operations
  - Laser cooling
  - Quantum state preparation and detection
- Laser stabilization schemes found in almost all atomic physics labs

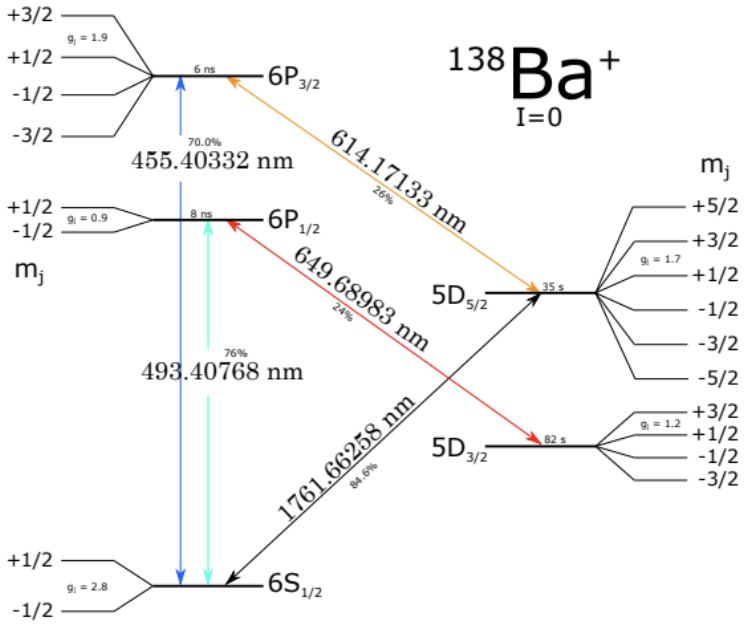
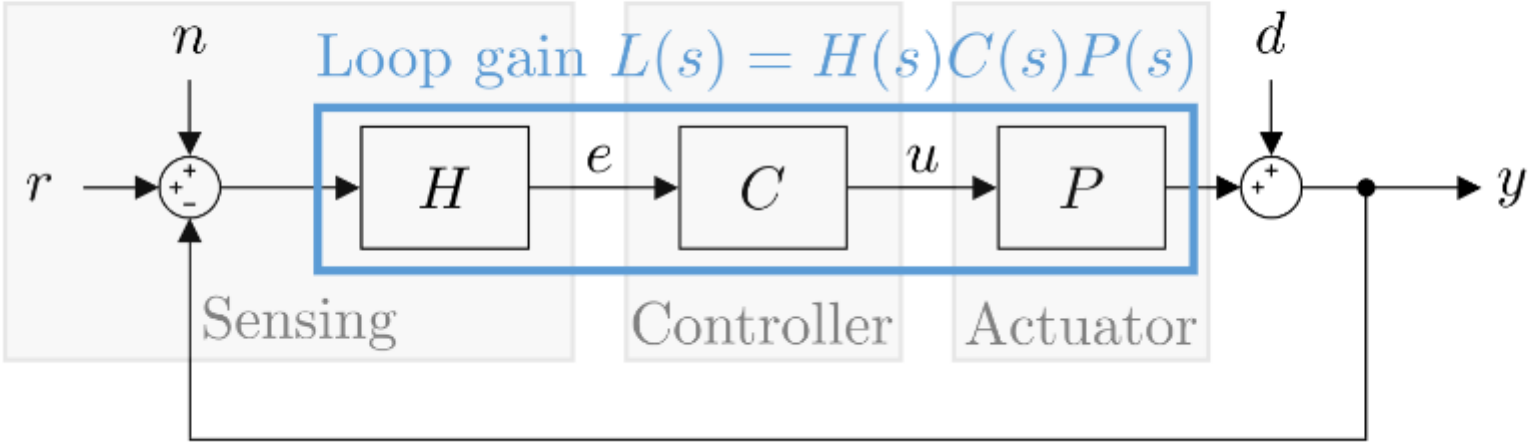


Figure 2.2: $^{138}Ba^{+}$ electronic energy structure.

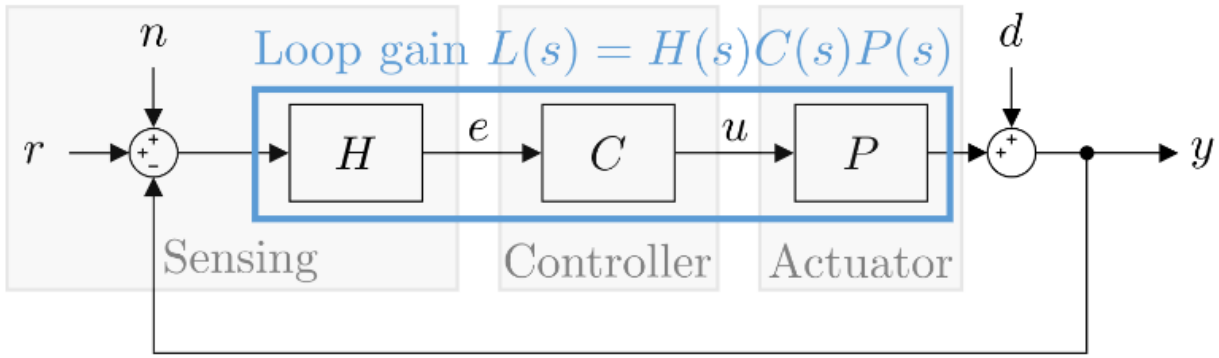
Feedback control

- Goal: make a system's output follow a desired reference



Feedback control of laser frequency

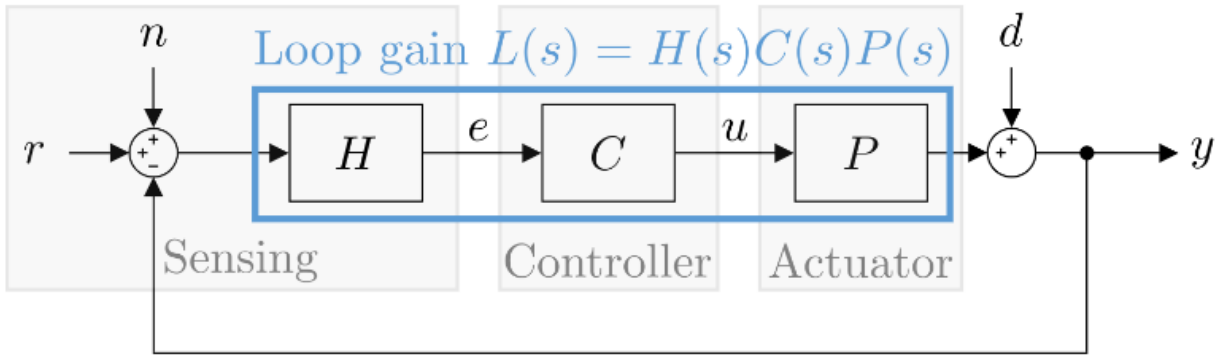
- Actuator/transducer
  - How to change the laser frequency?
- Controller
  - How to calculate the inputs to the actuator?
- Reference
  - Which frequency do we desire?
- Sensor
  - What is the difference between the reference and the actual frequency?



Feedback control of laser frequency

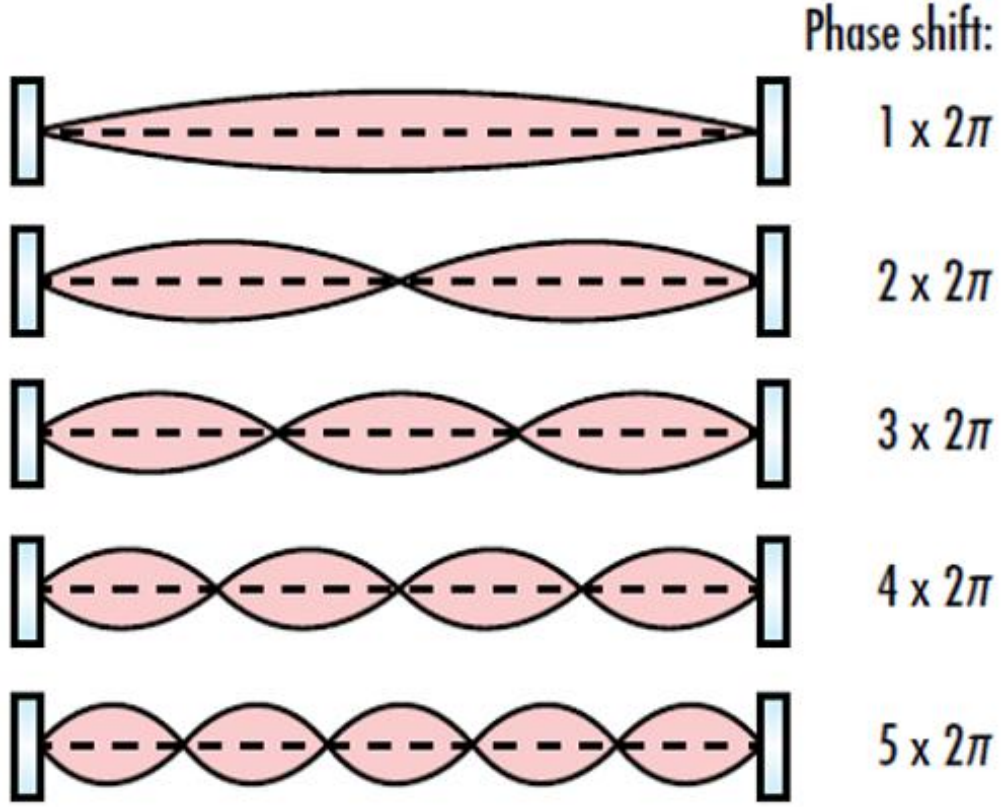
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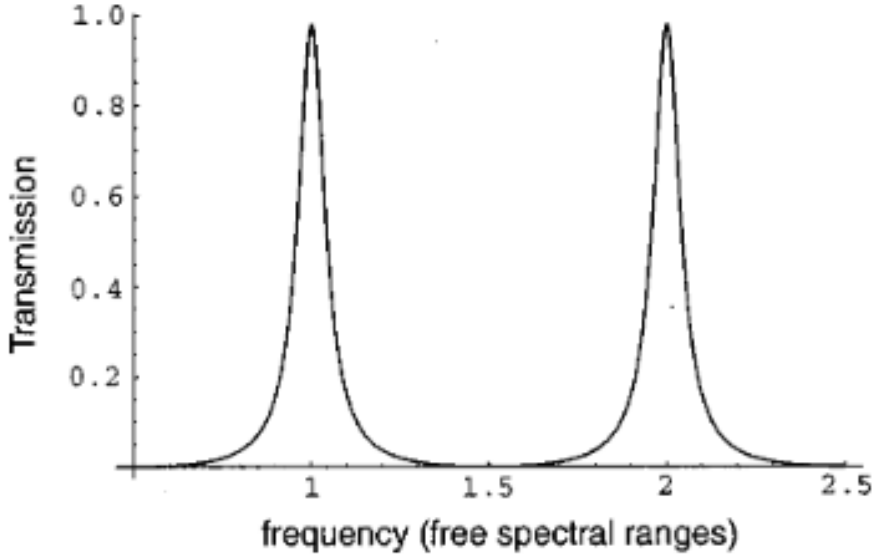


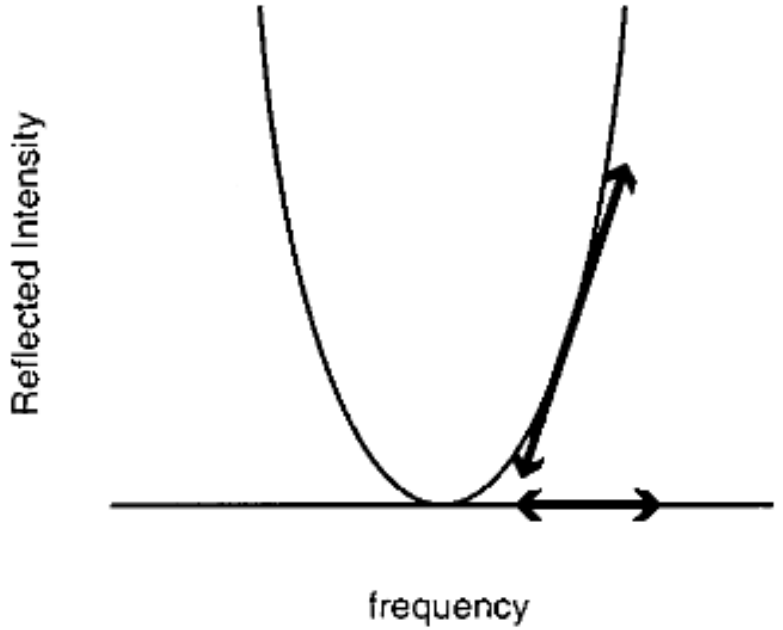
Optical cavity as a reference

- Provides a set of stable resonance modes spaced by $\nu_{\mathrm{FSR}} = \frac{\mathrm{c}}{2\mathrm{Ln}}$
  - L: Distance between the mirrors
  - n: Refractive index of the propagation medium
  - Both must be kept stable!
- Compare the laser frequency to a cavity mode



Pound-Drever-Hall technique





The transmission signal has sharp features at the resonances

The reflection signal is zero on resonance

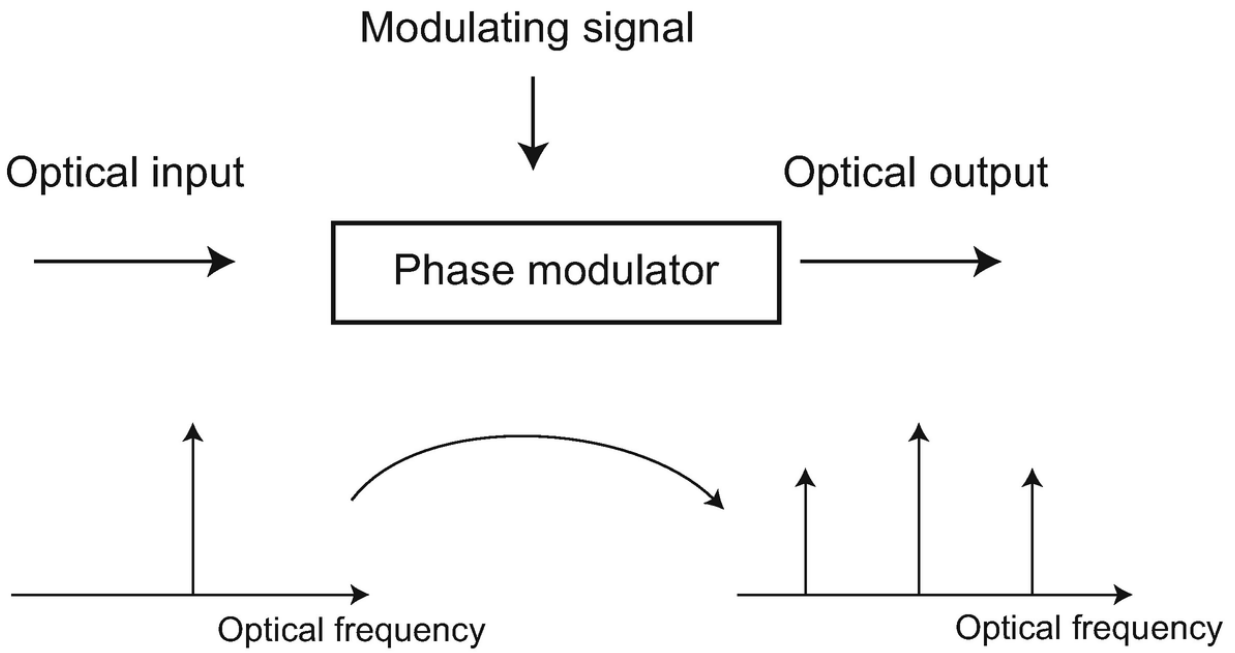
But it is symmetric!

Use its derivative instead as an **error signal**

Black, An introduction to Pound–Drever–Hall laser frequency stabilization (2000)

Pound-Drever-Hall technique

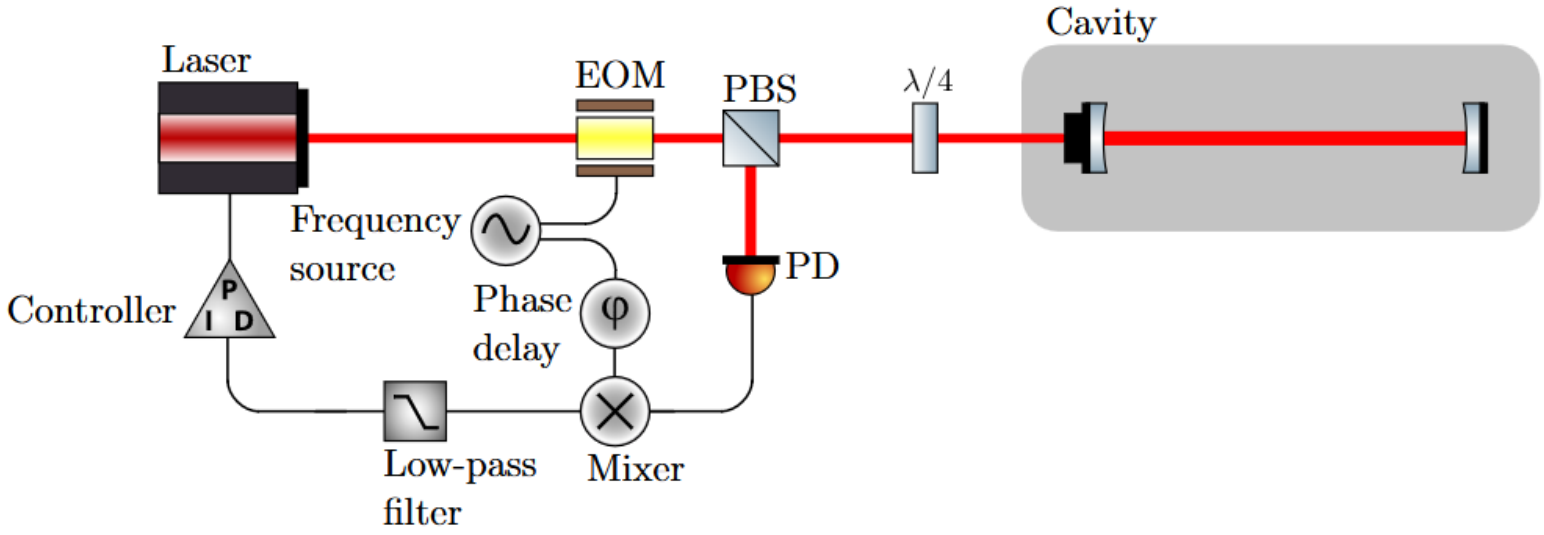
- How to access the derivative of the reflection signal?
  - Electro-optical modulator (EOM)
  - Imprints sidebands at the modulation frequency



Black, An introduction to Pound–Drever–Hall laser frequency stabilization (2000)

Pound-Drever-Hall technique

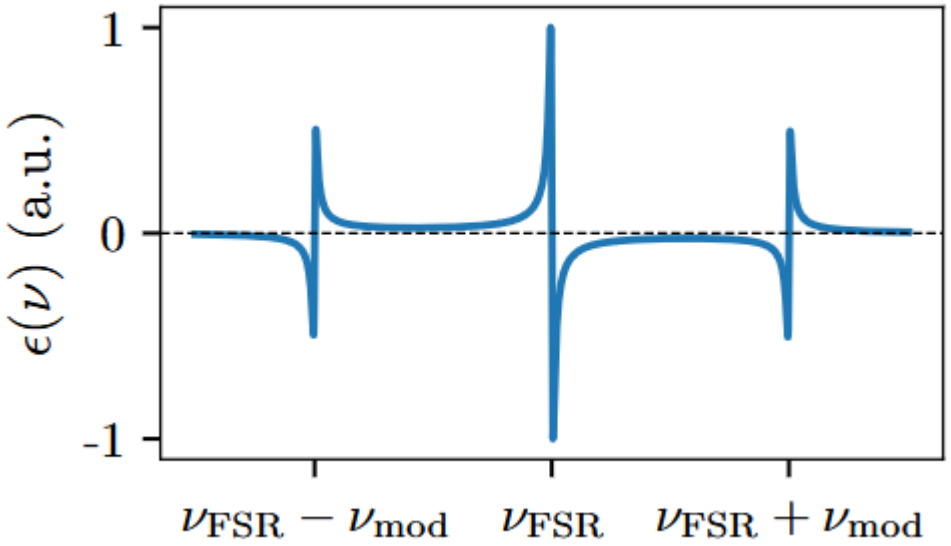
- The entire setup (details are not important)
- Important feature
  - Creates interference between the carrier and sidebands
  - Almost like taking the derivative

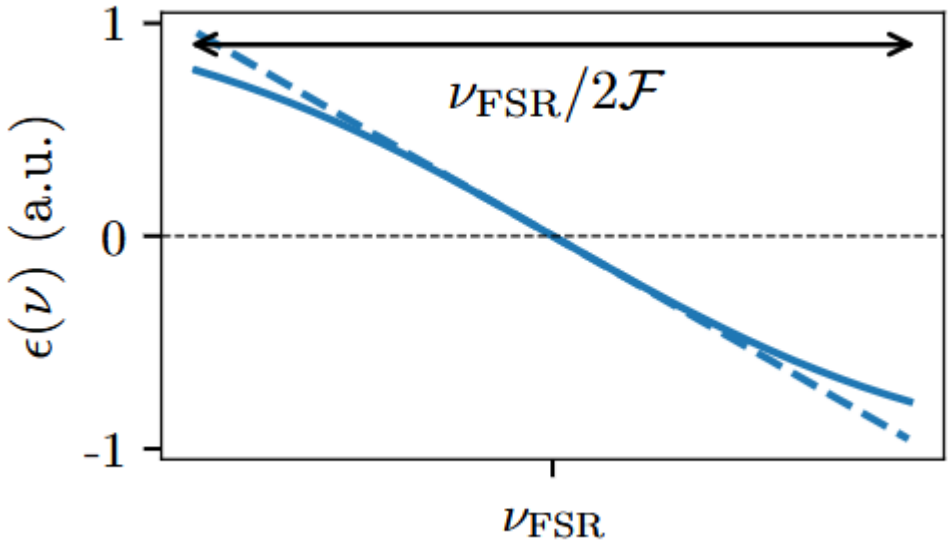


Black, An introduction to Pound–Drever–Hall laser frequency stabilization (2000)

Pound-Drever-Hall technique

- Error signal
  - Linear around the cavity resonance

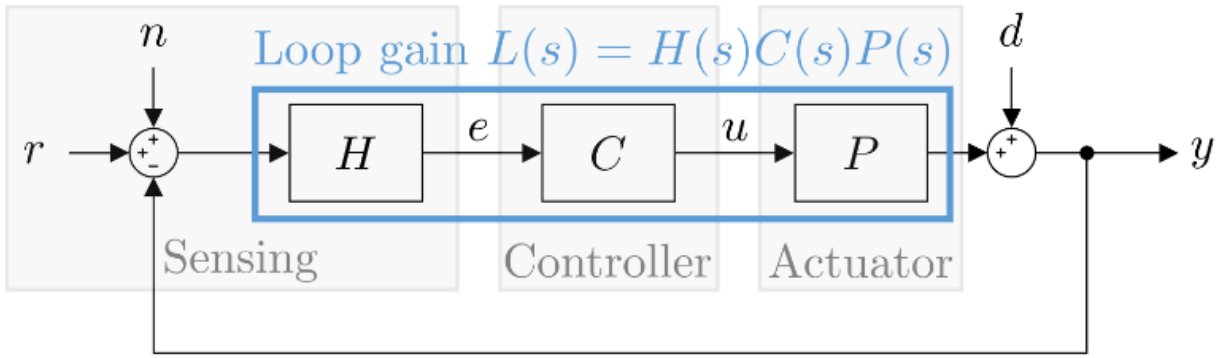




Black, An introduction to Pound–Drever–Hall laser frequency stabilization (2000)

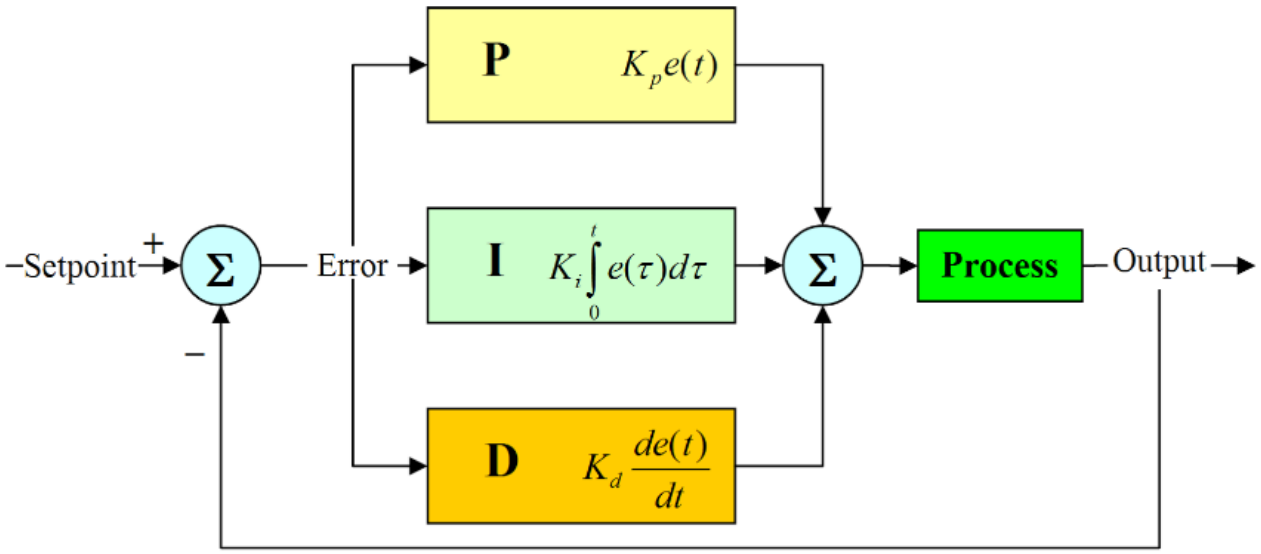
Feedback control of laser frequency

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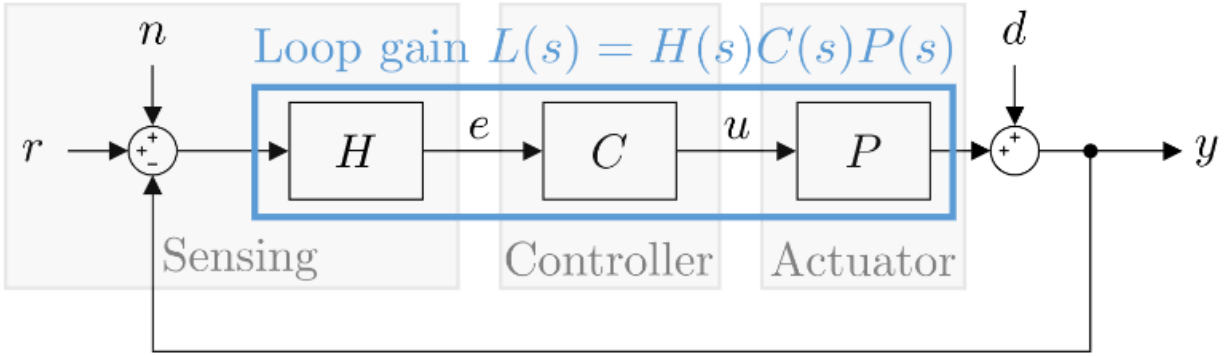
PID Controller

- **P**roportional gain:
  - Corrects for current deviations
- **I**ntegral gain:
  - Corrects for past deviations
- **D**ifferential gain:
  - Corrects for future deviations



Oswald, R. (2022). *Characterization and control of a cryogenic ion trap apparatus and laser systems for quantum computing* (Doctoral dissertation, ETH Zurich)

Transfer functions



Disturbance attenuation

$$S(s) = \frac{1}{1 + L(s)} = \frac{Y}{D}$$

Sensitivity function (S)

How well are disturbances rejected?

Reference following & Meas. noise injection

$$T(s) = \frac{L(s)}{1 + L(s)} = \frac{Y}{R} = \frac{Y}{N}$$

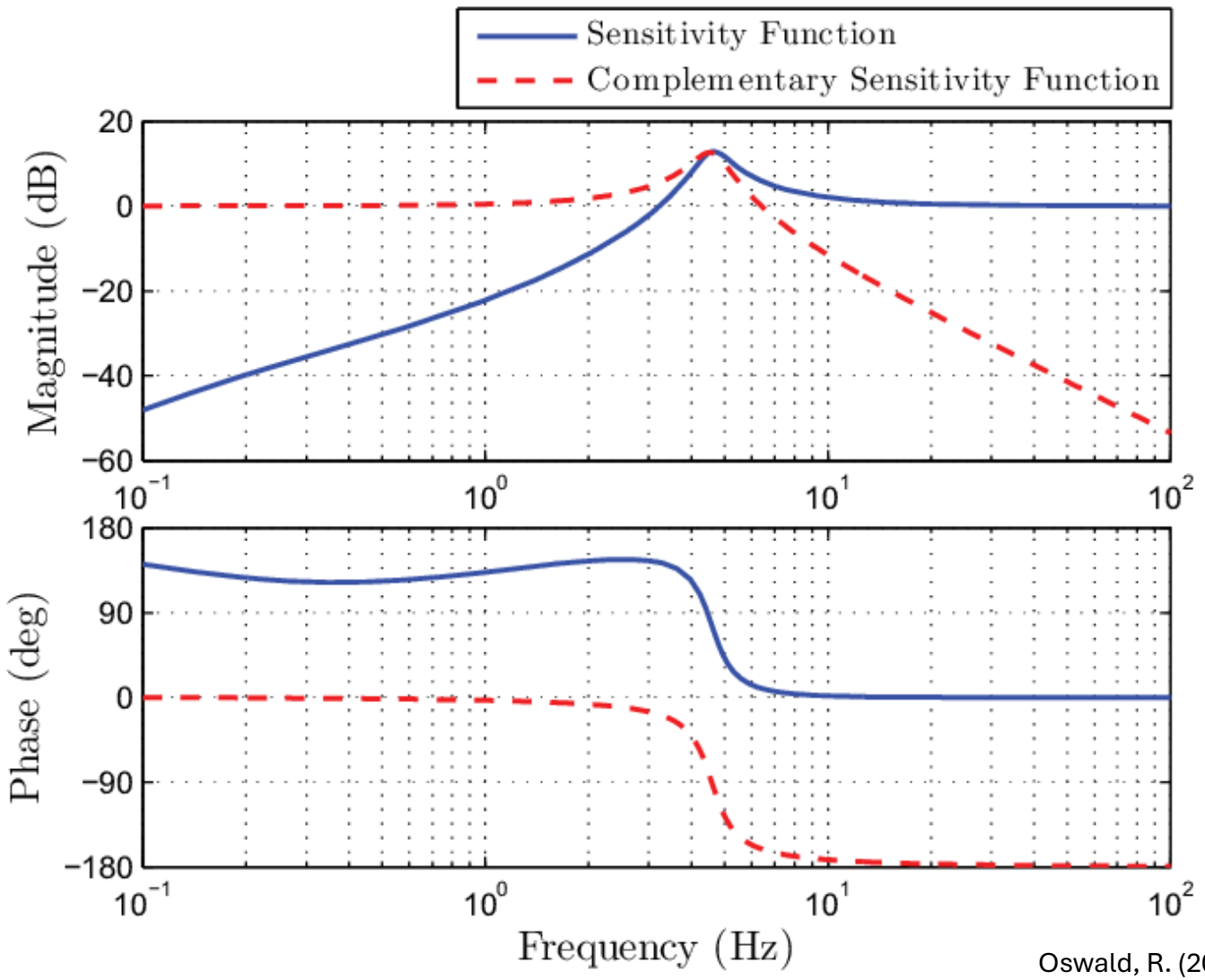
Complementary sensitivity function (T)

How well is the reference followed?

But also, how well is noise rejected?

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Control loop design



Bode plots

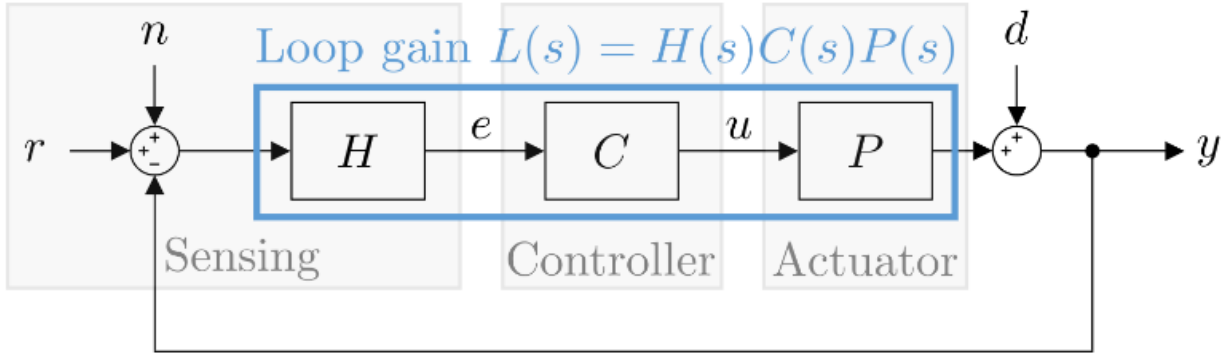
Controller performance in the Laplace domain (complex frequency domain)

Think: If I excite the system at a certain frequency, how will the system respond?

- At low frequencies:
  - $T(s) = 1$ and $S(s) = 0$
  - Excellent reference tracking and disturbance rejection
- At high frequencies:
  - $T(s) = 0$ and $S(s) = 1$
  - Control loop inactive to not amplify noise

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Control loop design



Disturbance attenuation

$$S(s) = \frac{1}{1 + L(s)} = \frac{Y}{D}$$

Reference following & Meas. noise injection

$$T(s) = \frac{L(s)}{1 + L(s)} = \frac{Y}{R} = \frac{Y}{N}$$

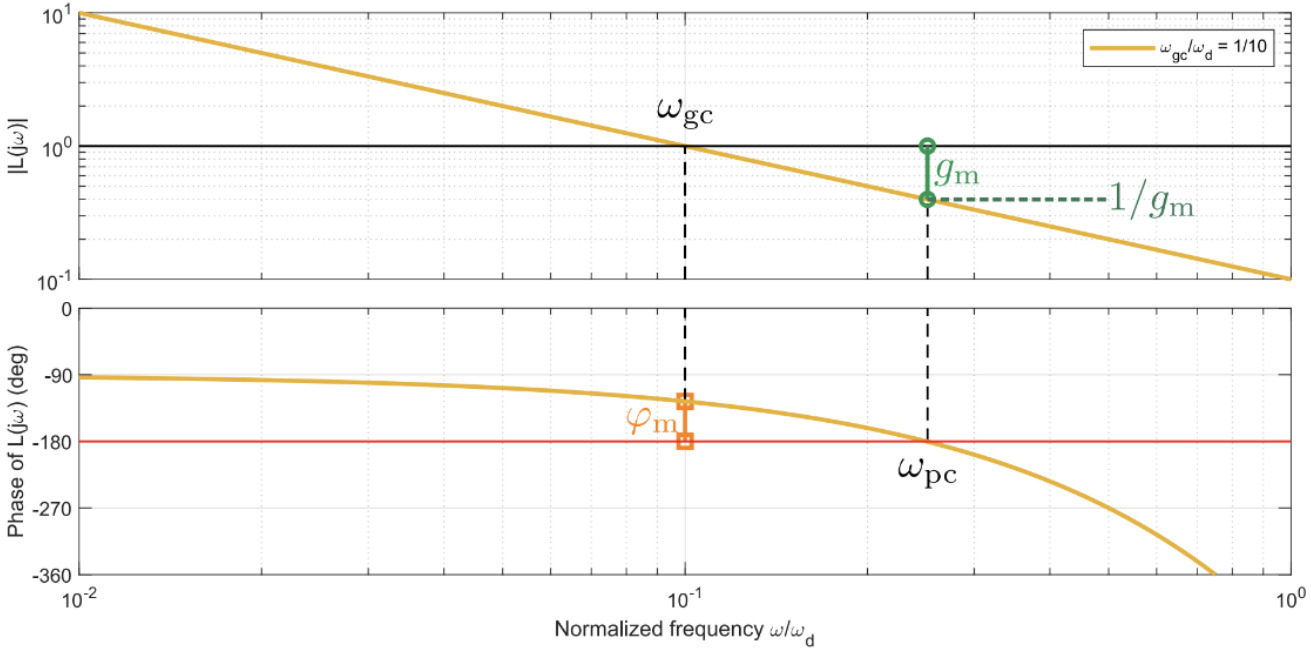
A loop gain in the form $L(s) \propto 1/s$ satisfies these requirements

Control engineering:

Choose P, I, and D parameters such that we achieve $L(s) \propto 1/s$

Oswald, R. (2022). *Characterization and control of a cryogenic ion trap apparatus and laser systems for quantum computing* (Doctoral dissertation, ETH Zurich)

Control bandwidth



$$L(s) = \frac{\omega_{gc}}{s} e^{-sT_d}$$

Disturbance attenuation

$$S(s) = \frac{1}{1 + L(s)} = \frac{Y}{D}$$

Reference following & Meas. noise injection

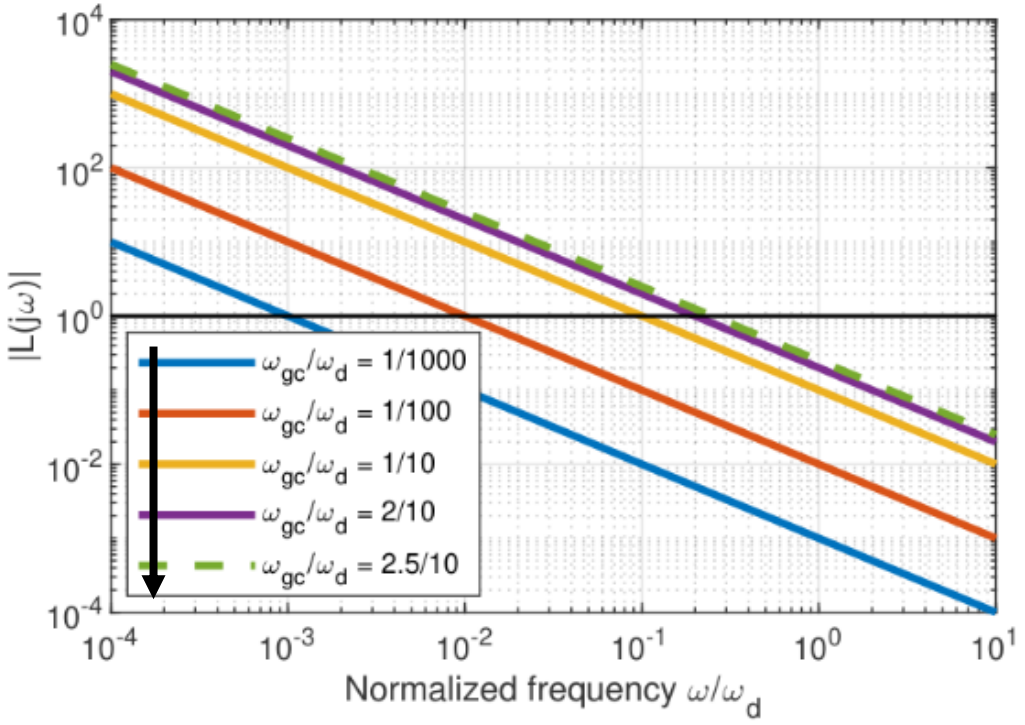
$$T(s) = \frac{L(s)}{1 + L(s)} = \frac{Y}{R} = \frac{Y}{N}$$

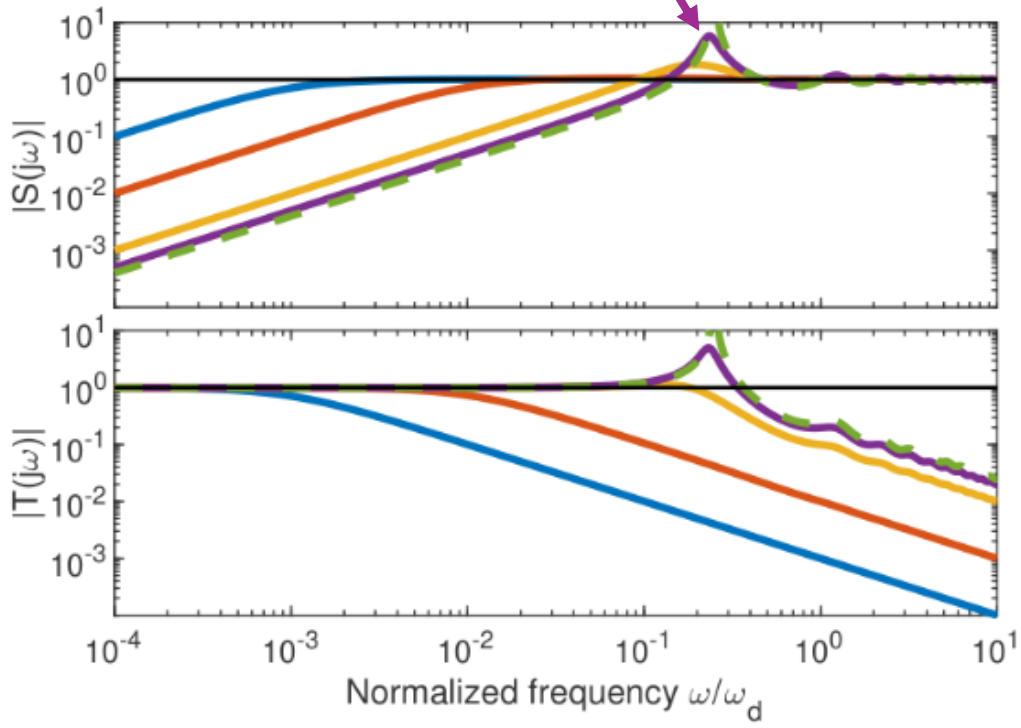
- The control loop is active as long as $|L(s)| > 1$
  - Sets the controller bandwidth
- We can increase this with the P-gain
  - $L(s) \propto k_p/s$
- But what happens if we push this too far?
  - Avoid $L(s) = -1$!

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Control bandwidth and the servo bump

Servo bump
- Estimate of the maximum achievable bandwidth



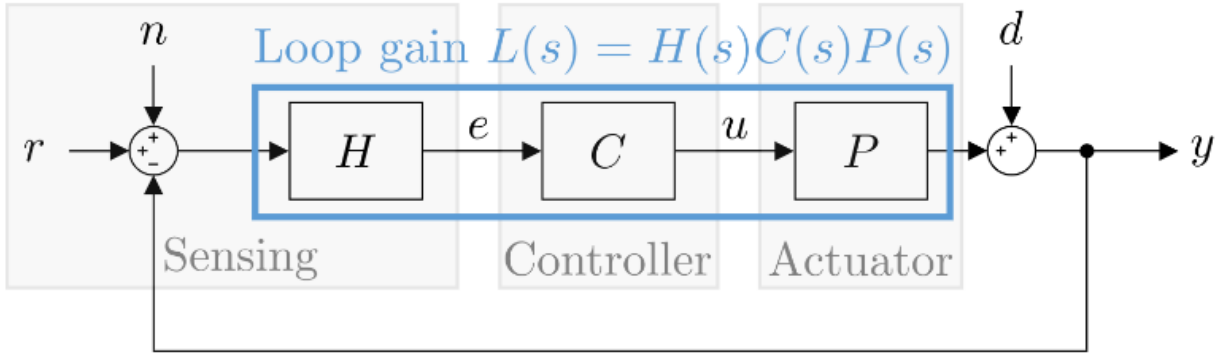


In the direction of the arrow – increased P-gain
- Increased bandwidth
- But also closer to instabilities $L(s) = -1$

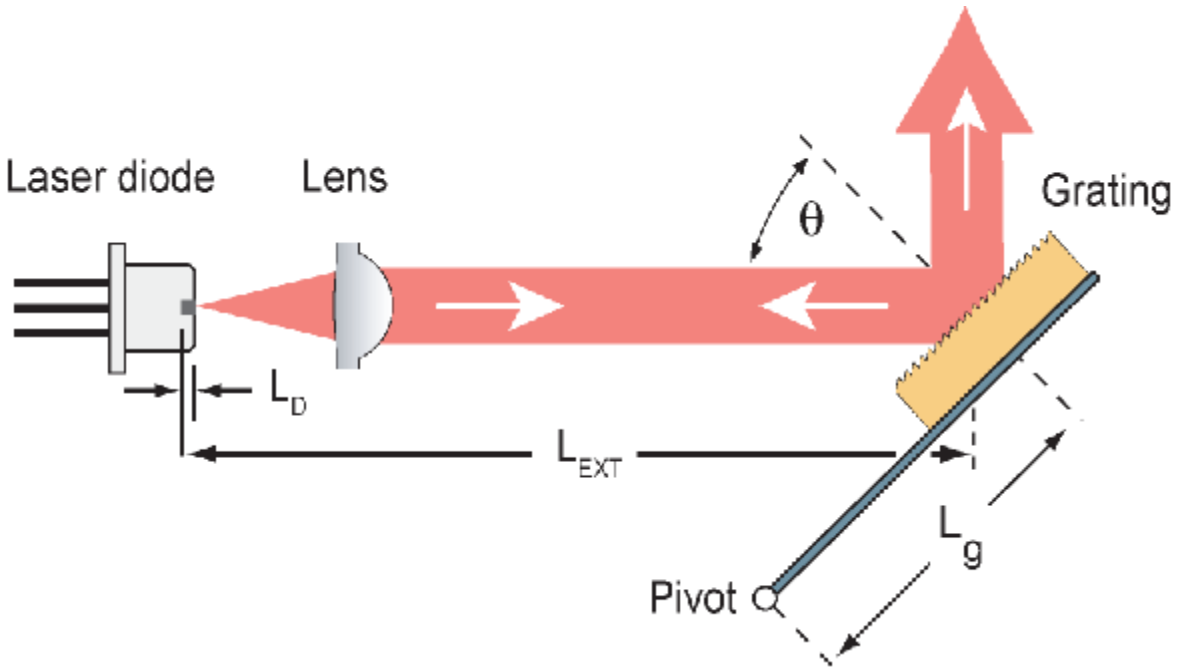
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Feedback control of laser frequency

- Actuator/transducer
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External cavity laser diode



Tune L_{EXT} using a piezo

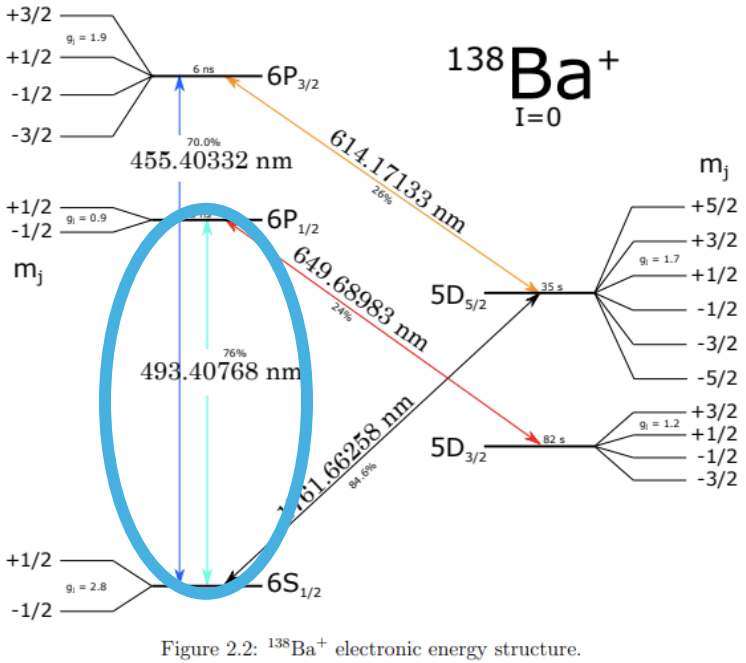


Figure 2.2: $^{138}Ba^{+}$ electronic energy structure.

Laser at 493 nm. Used for Doppler cooling and state detection of barium ions.

Note: This is a broad atomic transition, much broader than our laser linewidth.
We don't need to narrow down the laser, just to stop it from drifting around – easier to make the laser stabilization setup.

Tasks in the lab

1. Characterize the cavity used for laser frequency stabilization
2. Stabilize the laser to a cavity mode
3. Investigate the role of PID parameters
4. Measure the noise in the laser frequency